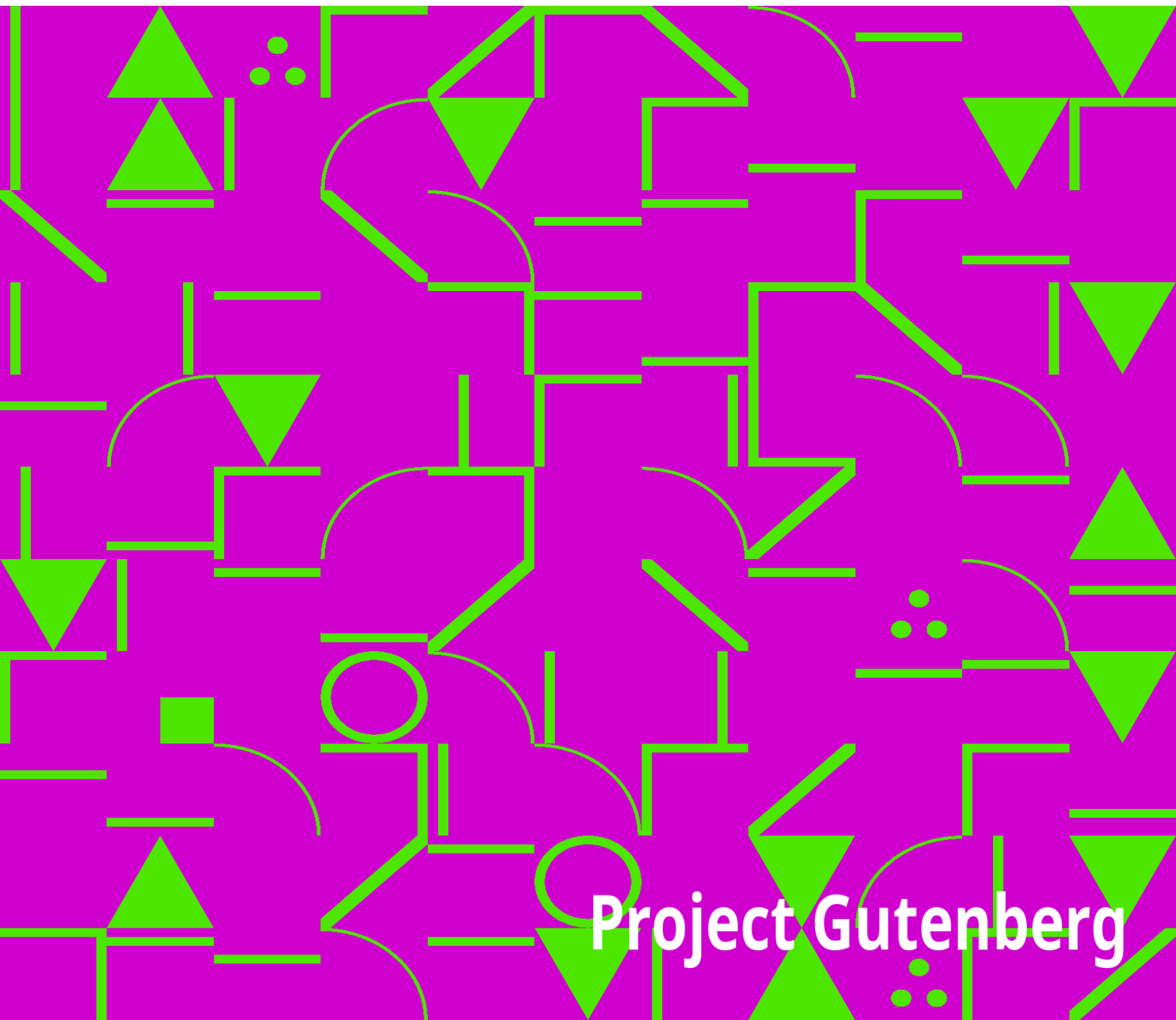


Taxonomic Status of Some Mice of The *Peromyscus boylii* Group in Eastern Mexico, With Description of a New Subspecies

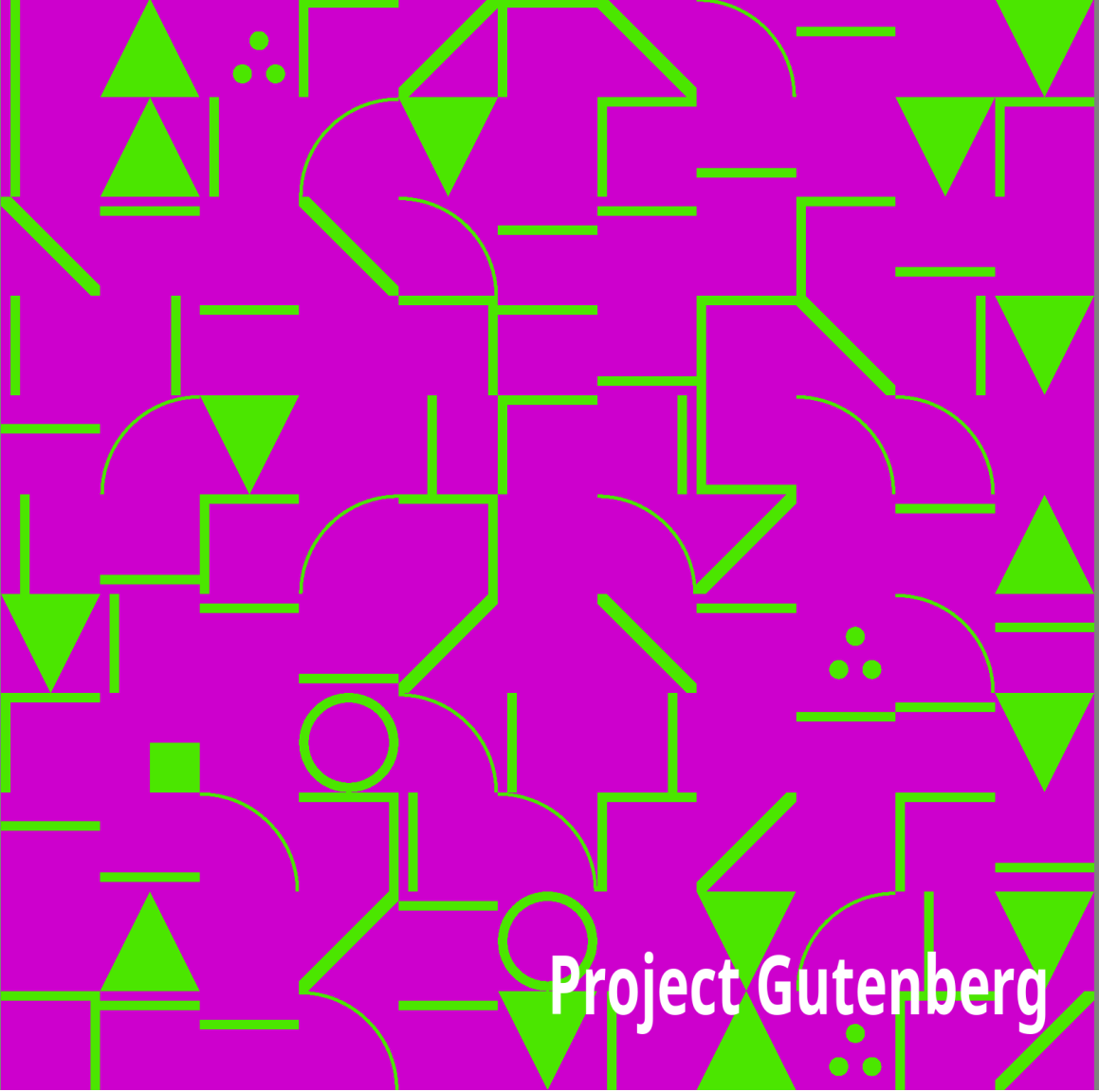
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The background of the lower half of the page is a vibrant, abstract pattern. It consists of a dense arrangement of various geometric shapes, including triangles, squares, circles, and lines, in shades of red, orange, and yellow. The shapes are scattered across the entire area, creating a complex and visually stimulating texture.

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*** START OF THE PROJECT GUTENBERG EBOOK TAXONOMIC STATUS OF SOME MICE OF THE PEROMYSCUS BOYLII GROUP IN EASTERN MEXICO, WITH DESCRIPTION OF A NEW SUBSPECIES ***

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UNIVERSITY OF KANSAS
LAWRENCE
1961

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BY
TICUL ALVAREZ

Saussure (1860) described *Peromyscus aztecus* from southern México. Osgood (1909) by comparison of one of Saussure's specimens with some from Mirador, Veracruz, concluded that *aztecus* was a subspecies of *P. boylii*. Dalquest (1953) incorrectly reported specimens of *P. boylii* from San Luis Potosí as *P. b. aztecus*. Merriam (1898) named *Peromyscus levipes* from Mt. Malinche, Tlaxcala. Thomas (1903) described from Orizaba, Veracruz, *P. beatae*, which Osgood (1909) mistakenly thought was indistinguishable from *P. boylii levipes*. Therefore, Osgood in 1909 in his revision of the genus *Peromyscus* reported only two subspecies of *P. boylii* from eastern México: *P. b. levipes*, and *P. b. aztecus*. Study of Osgood's and Thomas' material, along with recently collected specimens from the states of eastern México, leads me to conclude that *P. aztecus* and *P. boylii* are different species; that *P. beatae* is a valid subspecies different from *P. b. levipes*; and finally that specimens of *P. boylii* from Nuevo León and northwestern Tamaulipas pertain to an hitherto unnamed subspecies.

Peromyscus aztecus Saussure

1860. *H[esperomys]. aztecus* Saussure, Revue et Mag. Zool., Paris, ser. 2, 12:105, type from southern México, probably from the vicinity of Mirador, Veracruz, according to Osgood (N. Amer. Fauna, 28:156-157, April 17, 1909).

1909. *Peromyscus boylei aztecus*, Osgood, N. Amer. Fauna, 28:156, April 17.

Geographic distribution.—Known only from Mirador and Jalapa in Veracruz, and Huachinango in Puebla.

Diagnosis.—Size medium for the genus (see measurements); tail about as long as head and body; dorsal coloration near Sayal Brown (capitalized color terms after Ridgway, 1912); sides reddish; underparts Light Buff; tail bicolored but not distinctly so; supraorbital border of skull angular, and bullae pointed anteriorly; anterior half of braincase nearly straight (not rounded) as viewed from above; upper molar series long (4.7-5.0); incisive foramina short in relation to length of skull.

Comparisons.—From *Peromyscus boylii*, *P. aztecus* differs as follows: Larger in most parts measured; maxillary tooth-row 4.7-5.0 instead of 4.0-4.6; color brighter on sides (reddish instead of ochraceous); supraorbital border angular instead of rounded; anterior border of zygomatic plate convex in upper half and almost straight in lower half as opposed to nearly straight throughout in *boylii*; pterygoid fossa broader; bullae more pointed anteriorly and less inflated; mesostyles of upper molars larger; surface between orbital region and nasals convex in lateral view instead of flat.

Remarks.—When Saussure (1860:105) described *P. aztecus* he did not designate a type or type locality. Osgood (1909:157) designated as lectotype the mounted specimen, in the Geneva Museum, which has the skull inside and of which Saussure figured the molar teeth. Osgood (*loc. cit.*) examined one of the three specimens (No. 3926 USNM) that Saussure used in describing *P. aztecus* and found that it agreed "in every respect with recently collected specimens from Mirador, Veracruz, which, in the lack of exact knowledge, may be assumed to be the type locality, as it is certain that some at least of Saussure's specimens were taken near there."

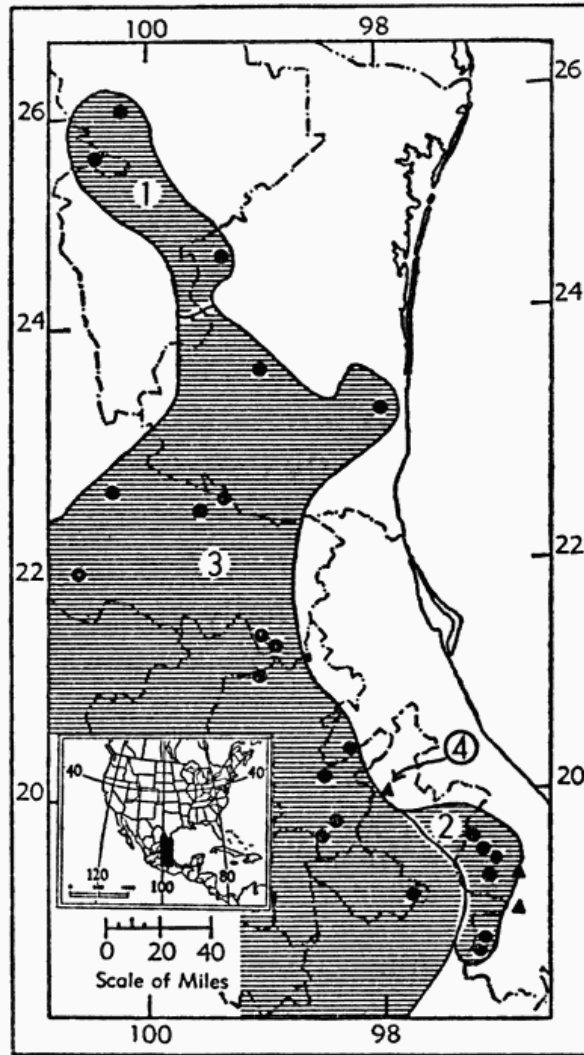


FIG. 1. Two species of *Peromyscus*.

1. *P. boylii ambiguus*
2. *P. boylii beatae*
3. *P. boylii levipes*
4. *P. aztecus* (triangles)

Osgood regarded *P. aztecus* as a subspecies of *P. boylii* because of the resemblance between *aztecus* and *P. b. evides*, but *evides* is far removed geographically (occurring only in western México) from *aztecus*, and is smaller. *P. aztecus* is larger than any known subspecies of *P. boylii*, and is not known to intergrade with *P. b. levipes* or *P. b. beatae* (with which *aztecus* occurs sympatrically at Jalapa, Veracruz), the two subspecies of *boylii* that are found nearest the geographic range of *P. aztecus*. Also, as mentioned previously, *aztecus* possesses distinctive characters that distinguish it from all subspecies of *boylii*. For these reasons I regard *aztecus* as a distinct species.

According to Hall and Kelson (1959:634), *P. aztecus* occurs in San Luis Potosí, Hidalgo, and west-central Veracruz, but their map 364 is based on the records of Osgood (1909:158) and Dalquest (1953:143). I have examined all the specimens reported by the two authors last named and find that those from San Luis Potosí are *P. boylii levipes*.

The diagnosis and comparisons here presented of *aztecus* were based on specimens from Mirador in comparison with all the specimens of *P. boylii* from eastern México listed beyond. The largest specimens of *P. boylii* that I have examined are from Las Vigas, Veracruz, and localities within a radius of five kilometers thereof. Some measurements of these large specimens of *P. boylii* overlap those of *P. aztecus* but the two kinds of mice differ greatly in characters of the skull, in color, and in length of tail.

The specimens (three adults and three juveniles) from Huachinango, Puebla, are slightly darker than specimens from Mirador but do not differ otherwise. Of two specimens reported from Jalapa, Veracruz, by Osgood (1909:158), one (108547 USNM) agrees with specimens from Mirador in color and cranial characteristics and is *P. aztecus*, whereas the other (108548 USNM) is *P. b. beatae*.

Specimens examined.—Total 16 (all USNM) as follows: PUEBLA: Huachinango, 6. VERACRUZ: Mirador, 9; Jalapa, 1.

***Peromyscus boylii levipes* Merriam**

1898. *Peromyscus levipes* Merriam, Proc. Biol. Soc. Washington, 12:123, April 30, type from Mt. Malinche, 8400 ft., Tlaxcala.

1909. *Peromyscus boylei levipes*, Osgood, N. Amer. Fauna, 28:153, April 17.

Geographic distribution.—Southeastern Tamaulipas and eastern San Luis Potosí, south through the central states of México to Guatemala.

Diagnosis.—Size medium for the species; tail shorter or longer than head and body (83-112.3%); color variable according to locality but in general ochraceous, having some dusky on upper parts; supraorbital border not angular, almost rounded; auditory bullae large.

Comparisons.—For comparisons see accounts of the subspecies discussed beyond and Osgood (1909:145).

Remarks.—A precise diagnosis for *P. b. levipes* is difficult to prepare because some geographic variation in color and in the cranial characters is present within the range of the subspecies as here understood. For instance there is a gradual cline of decreasing size to the northward in nearly all measurements, but the ratio of length of tail to length of head and body does not present such a cline; mice from several localities in San Luis Potosí have a relatively shorter tail than do mice from farther north and from farther south. Also, specimens labeled in reference to Zacualpilla, Jacales, Jacala, Tulancingo, and San Miguel Regla average slightly darker dorsally than do typotypes. Some of these specimens are reddish on the cheek and lateral line. Specimens from San Luis Potosí resemble topotypes, but some specimens from northeastern localities in that state have cinnamon or brownish upper parts and are intermediate in coloration between populations of *levipes* to the south and populations of the same subspecies to the north from the Sierra Madre Oriental and the Sierra de Tamaulipas. Specimens from these two sierras have a cinnamon-reddish color that is more intense in specimens from the Sierra de Tamaulipas.

Osgood (1909:153) recorded *P. b. levipes* as occurring from central Nuevo León south through San Luis Potosí, Hidalgo, and Veracruz to southern Oaxaca. Actually specimens from Nuevo León and from most parts of Veracruz differ subspecifically from *levipes* and also from each other. In Veracruz, *P. b. levipes* is known only from the northwestern part.

Specimens examined.—Total 179 as follows: TAMAULIPAS: Sierra Madre Oriental, 5 mi. S, 3 mi. W Victoria, 1900 ft., 2; 8 mi. S, 6 mi. W Victoria, 4000 ft., 37; Sierra de Tamaulipas, 2000 ft., 8 mi. S, 11 mi. W Piedra, 13. SAN LUIS POTOSÍ: Villar, 11 (USNM); 10 km. E Platanito, 19 (LSU); 8 mi. E (by road) Santa Barbarita, 12 (LSU); Agua Zarca, 3 (LSU); 6 km. NE Cd. Maíz, 13 (LSU); Pendencia Region (Puerto Lobos), 1 (LSU); Pendencia, 2-1/2 mi. N Puerto Lobos, 5 (LSU); 3 km. SW San Isidro, 15 (LSU); Cerro Coneja Region, Llano Coneja, 6100 ft., 2 (LSU); Xilitla, 4 (LSU). HIDALGO: 10 mi. NE Jacala, 5050 ft., 7; Regla (San Miguel), 2250 m., 4; Arroyo de las Tinajas, 2370 m., 9.5 km. SSW Tulancingo, 1; 10 mi. NW Apam, 7750 ft., 1. VERACRUZ: 3 km. N Zacualpan, 6000 ft., 1; 3 km. W Zacualpan, 6000 ft., 12; 2 km. N Los Jacales, 7500 ft., 8; 6 km. WSW Zacualpilla, 6500 ft., 5. TLAXCALA: Mt. Malinche, 3 (USNM).

***Peromyscus boylii beatae* Thomas**

1903. *Peromyscus beatae* Thomas, Ann. Mag. Nat. Hist., ser. 7, 11:485, May, type from Xometla Camp, Mt. Orizaba, Veracruz.

Geographic distribution.—East side of the Sierra Madre Oriental in Veracruz, from Jalancingo south to Xuchil.

Diagnosis.—Size large for the species; tail no shorter than head and body (100-114.8%); dorsum dark (near Prout's Brown or Mummy Brown middorsally, Clay Color on sides); supraorbital border rounded; anterior palatine foramina long.

Comparisons.—*P. b. beatae* differs from other subspecies of *P. boylii* by the combination of large size, long tail, and dark color.

Remarks.—Thomas (1903:485) described *P. beatae* on the basis of five specimens from Xometla Camp (lat. 18° 59' N, long. 97° 10' W) and one juvenile from Santa Barbara Camp, both on the Volcán de Orizaba, Veracruz. Thomas thought that *beatae* was related to *aztecus*, but the differences relied on by him to distinguish the two are the same as those that distinguish *aztecus* from *boylii*. Osgood (1909:153) placed *beatae* in synonymy under *P. b. levipes* because Mount Orizaba (type locality of *beatae*) is "relatively very near" Mount Malinche (type locality of *levipes*), and Thomas had not compared *beatae* with *levipes*. Xometla, on the east side of the Volcán de Orizaba, is approximately 56 miles east of the Tlaxcalan part of Mount Malinche and is situated where the Tropical Life-zone begins, whereas Mount Malinche is in the Austral Life-zone on the Mexican Plateau; the difference in habitat between the two places is great. Topotypes of *levipes* differ from two topotypes of *beatae* in the same fashion as do other specimens of *levipes* (from San Luis Potosí) from other specimens of *beatae* (from Veracruz). Unfortunately, the topotypes of *beatae* lack external measurements and are subadults, but their coloration agrees with that of other specimens that are here referred to *beatae*.

Hall and Kelson (1959:634, map 364) incorrectly mapped the distribution of *levipes* in Veracruz. There are at least two places named Xuchil in the state of Veracruz and Hall and Kelson (*loc. cit.*) unfortunately plotted the one at lat. 20° 42' N, long. 97° 42' W whereas the specimens actually were collected at the Xuchil on the plateau south of the Volcán de Orizaba (18° 53' N, 97° 14' W) in the west-central part of Veracruz. The specimens from Xuchil are *P. b. beatae*.

Intergradation in color between the two subspecies *levipes* and *beatae* is seen in specimens from Jalapa and Zacualpan (3 km. N, also others from 3 km. W), Veracruz. Intergradation between these two subspecies possibly will be found elsewhere along the Sierra Madre Oriental.

Specimens examined.—Total 60 as follows: VERACRUZ: 1 km. E Jalancingo, 6500 ft., 2; 2 km. S Jalancingo, 2; 6 km. SSE Altotonga, 8000 ft., 8; 1 km. W Las Vigas, 8500 ft., 2; Las Vigas, 8500 ft., 13; 2 km. E Las Vigas, 8000 ft., 5; 3 km. E Las Vigas, 8000 ft., 8; 5 km. E Las Vigas, 7 (TAM); 5 km. N Jalapa, 4500 ft., 2; Jalapa, 1 (USNM); 10 km. SE Perote, N slope Cofre de Perote, 10,500 ft., 1 (TAM); Xometla Camp, Mt. Orizaba, 8500 ft., 2 (BM); Sta. Barbara, Mt. Orizaba, 12,000 ft., 1 (BM); Xuchil, 6 (CM).

***Peromyscus boylii ambiguus* new subspecies**

Type.—Male, adult, skin and skull, No. 33092, United States National Museum, from Monterrey, Nuevo León; obtained on February 17, 1891, by Wm. Lloyd, original number 377.

Geographic distribution.—Eastern Coahuila, central Nuevo León, and the Sierra San Carlos, Tamaulipas.

Diagnosis.—Size small for the species; tail averaging longer than head and body (90-114%); dorsal coloration ochraceous, slightly darker middorsally;

cheeks and lateral line Capucine Orange; skull small; supraorbital border rounded; anterior palatine foramina short.

Comparisons.—*P. b. ambiguus* differs from *P. b. levipes* in smaller size, longer tail relative to length of head and body, smaller incisive foramina, brighter and paler color, and relatively broader interorbital region. From *P. b. beatae*, *P. b. ambiguus* differs in being smaller in all parts measured and paler.

Remarks.—Osgood (1909:155) reported as *P. b. levipes* 37 specimens from Monterrey and 18 from Cerro de la Silla, Nuevo León, but noted that they were "aberrant." I have examined those same specimens and can hardly decide to which species, *P. boylii* or *P. pectoralis*, they belong. Everything considered I, as did Osgood, opine that the specimens are *P. boylii*. However, I do not rule out the

possibility that in this area there is an unnamed species, because I find an unusually wide range of variation in such cranial characters as size of the bullae, width and form of the pterygoid fossa, and shape of the braincase. Extremes of these characters are not constantly associated except in one specimen (33124 USNM), which is the smallest of all the adults examined. It has small bullae, a short rostrum, widely spreading zygomatic arches anteriorly, and a narrow pterygoid fossa, but does not differ externally from the other specimens. Additional material from this area is needed in order to make out the systematic position of these mice.

The specimens from the Sierra San Carlos, Tamaulipas, closely resemble *levipes* in color, but are referred to *ambiguus* on the basis of small size, as also are the two specimens from 12 km. E San Antonio de las Alazanas, Coahuila.

[illegible]

11 mean	205.6	99.5	22.4	93.8	28.5	14.2	4.4	11.3	5.9	4.4
max.	219	114	23	108.6	30.5	14.5	4.6	12.2	6.4	4.6
min.	193	90	21	87.4	27.2	13.8	4.2	10.8	5.6	4.1
6 km. NE Cd. del Maíz, San Luis Potosí										
9 mean	198.7	96	22	93.4	28.1	14.0	4.4	11.2	6.0	4.5
max.	205	105	22	105.0	28.7	14.2	4.6	11.7	6.4	4.6
min.	187	90	22	85.7	27.3	13.4	4.3	10.6	5.7	4.3
11 mi. W, 8 mi. S Piedra, Tamaulipas										
5 mean	201.8	101.8	22.6	101.8	28.5	14.0	4.4	11.3	6.1	4.3
max.	214	110	23	109.3	29.0	14.1	4.6	11.5	6.2	4.7
min.	193	94	22	94.9	28.2	13.9	4.2	11.0	6.0	4.1
<i>P. boylii ambiguus</i> La Vegonia, Tamaulipas										
7 mean	199.1	101.6	21.3	104.3	26.9	13.4	4.3	10.5	5.6	4.3
max.	213	109	22.4	108.9	28.6	13.7	4.5	11.8	5.9	4.7
min.	188	97	20	98.0	26.4	13.2	4.2	9.5	5.3	4.1
Monterrey, Nuevo León										
16 mean	199.7	103.2	21.3	106.9	27.6	13.9	4.4	10.7	5.6	4.2
max.	216	114	22	125.6	28.2	14.9	4.6	11.5	5.8	4.5
min.	176	92	19	88.0	26.8	13.2	4.1	10.2	5.0	4.0

Specimens examined.—Total 64 as follows: NUEVO LEÓN (USNM): Monterrey, 37; *Cerro de la Silla*, 18. COAHUILA: 12 km. E San Antonio de las Alazanas, 9000 ft., 2. TAMAULIPAS: La Vegonia, Sierra San Carlos, 7 (UMMZ).

I am grateful to the following persons for the loan of specimens: G. B. Corbet, British Museum, Natural History (BM); David H. Johnson, United States National Museum (USNM); George H. Lowery, Jr., Louisiana State University (LSU); Philip Hershkovitz, Chicago Natural History Museum (CM); William B. Davis, Texas Agricultural and Mechanical College (TAM); W. H. Burt and Emmet T. Hooper, University of Michigan Museum of Zoology (UMMZ). Specimens lacking designation as to collection are housed in the Museum of Natural History of The University of Kansas. I am indebted to Professor E. Raymond Hall and Mr. J. Knox Jones, Jr. for the use of these specimens and for other assistance. It is appropriate to record also that the findings reported above are an outgrowth of related work done as a Research Assistant under Grant No. 56 G 103 from the National Science Foundation.

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Transmitted June 30, 1961.

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Transcriber's Note

The following typographical errors are noted, but not corrected in the text:

page 116: "typotypes" should be "topotypes"
page 116: "Potosi" should be "Potosí"
page 116: "Sán Miguel" should be "San Miguel"
page 116: "Sán Isidro" should be "San Isidro"
page 117: "pleateau" should be "plateau"
page 120: "Nuevo Leon" should be "Nuevo León"

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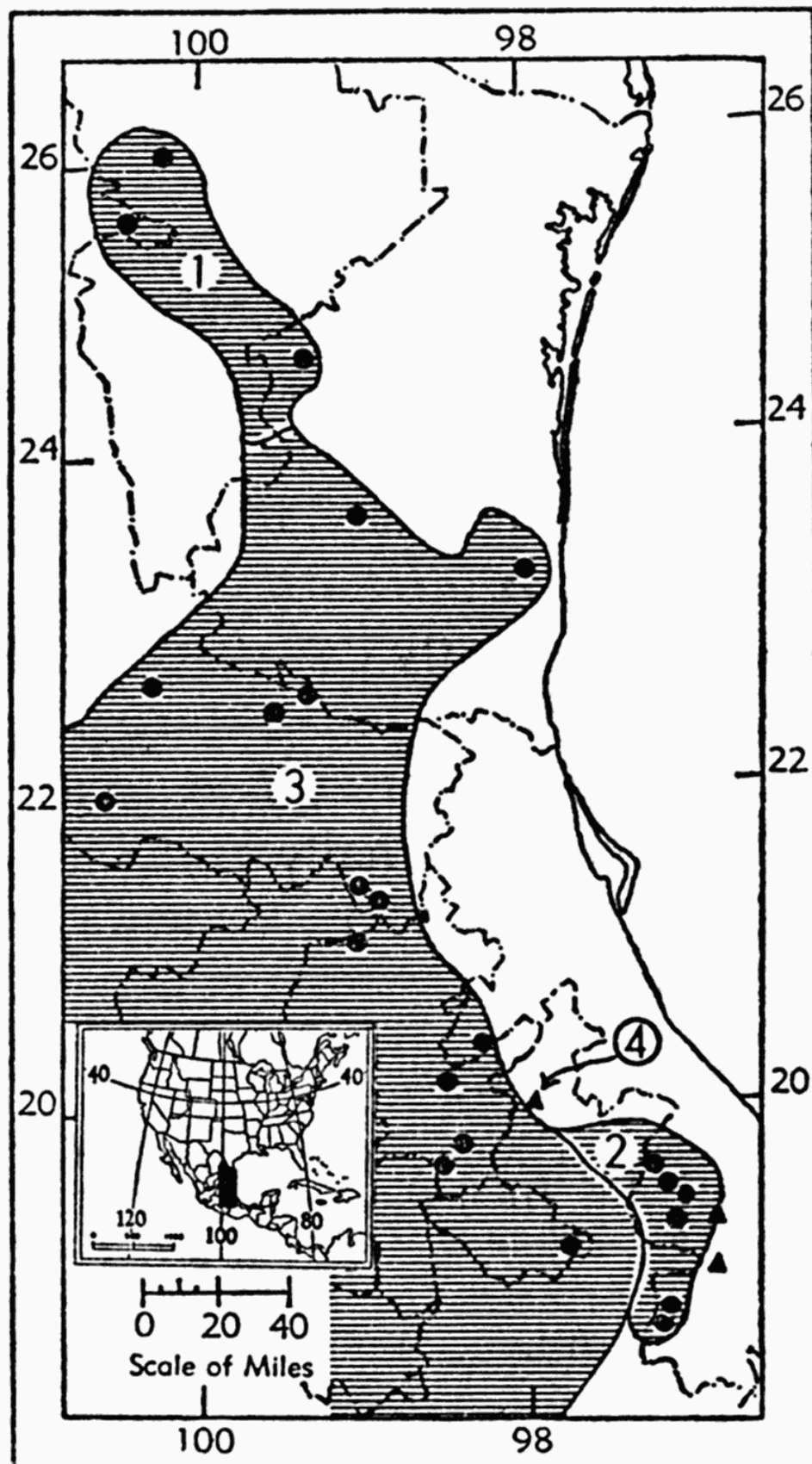
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